

Key features of quadratic functions



1

a 75

b 75

2

a 4

b 273

c 420

d 0

3

a Yes

b No

c Yes

d No

e Yes

f No

g No

h No

i No

4

a No

b Yes

c No

d Yes

e No

f No

g Yes

h Yes

i Yes

j No

k Yes

l No

m No

n No

o Yes

5

a No

b Yes

c No

d No

6

a Quadratic

b Linear

c Quadratic

d Neither

e Neither

7

a i $(4, 0), (-2, 0)$ ii $4, -2$

b i $(2, 0), (-4, 0)$ ii $2, -4$

c i $(3, 0)$ ii 3

8



a

i $-6, 2$

ii -12

iii Minimum value, -16

b

i $0, 4$

ii 0

iii Maximum value, 4

c

i 3

ii 9

iii Minimum value, 0

9

a -5

b -25

c 0

d $x < -5$

10

a $(-4, 2)$

b Minimum

c 2

11

a Minimum point

b

i No

ii Yes

iii No

iv Yes

12

a Maximum point.

b

i Yes

ii No

iii No

iv Yes

13 $x < 2$

14

a

x	-3	-2	-1	0	1	2	3
y	5	0	-3	-4	-3	0	5

b $2, -2$

c $(0, -4)$

d Minimum

e Positive

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- a
- i $(3, 2)$ ii Maximum iii 2
- b
- i $(4, 5)$ ii Minimum iii 5
- c
- i $(-4, -3)$ ii Maximum iii -3

16

- a Quadratic
- b 9
- c The height from which the soccer ball is kicked
- d 3
- e The time at which the soccer ball hits the ground

17

- a
- i Minimum, -4 ii Domain: all real x ,
Range: $y \geq -4$
- b
- i Maximum, 16 ii Domain: all real x
Range: $y \leq 16$
- c
- i Minimum, 0 ii Domain: $x > 7$
Range: $y \geq 0$
- d
- i Minimum, 0 ii Domain: $x < -4$
Range: $y \geq 0$

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- a
- i Minimum, -1 ii $y \geq -1$
- b
- i Maximum, -2 ii $y \leq -2$
- c
- i Minimum, -3 ii $y \geq -3$
- d
- i Maximum, -3 ii $y \leq -3$

